BT-203 (GS)

B.Tech., I & II Semester Examination, November 2022

Grading System (GS)

BASIC MECHANICAL ENGINEERING

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Full paper (Q1–Q8)

1. (a) Explain the methods to measure the hardness of materials. **7**

(b) A flat plate of area 1.5×10^6 mm² is pulled with a speed of 0.4 m/s relative to another plate located at a distance of 0.15 mm from it. Find the force and power required to maintain this speed, if the fluid separating them has viscosity 1 poise. [1 poise = 0.1 N·s/m²] **7**

2. (a) Classify engineering materials. What do you mean by alloys? Write some applications of alloys. **7**

(b) A rod 200 cm long and of diameter 3.0 cm is subjected to an axial pull of 30 kN. If the Young's modulus of the material of the rod is 2×10^5 N/mm², determine: (i) Stress (ii) Strain (iii) Elongation of the rod. **7**

3. (a) With a neat diagram explain the main parts of a lathe machine. Explain any two operations performed on a lathe machine. **7**

(b) During a trial operation of a boiler, 1630 kg of coal were consumed in 24 hrs and 13000 kg of water evaporated. The mean steam pressure was 7 kgf/cm² and condition dry saturated. The fuel contained 3 percent moisture and 4 percent ash and had a calorific value of 7200 kcal/kg. The feed water temperature was 35°C. Determine: (i) boiler efficiency (ii) equivalent evaporation per kg of dry coal (iii) equivalent evaporation per kg of combustibles. **7**

4. (a) Explain the working of a manometer for pressure measurement. Also relate absolute pressure, gauge pressure and vacuum pressure. **7**

(b) Explain the measurement process using vernier calliper with neat sketch. **7**

5. (a) Define Viscosity. What is the importance of viscosity while selecting fluids in various engineering applications? **7**

(b) Discuss briefly the differences between externally fired and internally fired steam generators. Also discuss the application of each type of steam generator. **7**

6. (a) Explain working principle of hydraulic turbines (Pelton wheel) with neat sketch. **7**

(b) Derive an expression for efficiency of Otto cycle with neat sketches of P–V and T–S diagrams. **7**

7. (a) State the Zeroth Law of thermodynamics. How is mercury in a thermometer able to find the temperature of a body using the Zeroth law of thermodynamics? **7**

(b) Explain the working of four stroke diesel engine with neat sketch. **7**

8. Write short notes on any two: **14**

(a) Welding defects (b) Dial gauge (c) Sine bar (d) Thermodynamic properties

****

Yeh reconstructed paper kya hai? RGPV ke 2022–2026 ke papers ka free official PDF internet par downloadable nahi hai (mirrors sirf ~2019 tak ke papers rakhte hain). Ye version un hi papers ke public web renders se word-by-word English questions lekar banaya gaya hai, tai ki students unhe PDF me padh/print kar sakein. Hindi version aur exact layout ke liye original paper dekhein. Ye unofficial study material hai.